

# ITA0407 - STATISTICS WITH R PROGRAMMING

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## 1. Arithmetic Operations

**Explanation:** Arithmetic operators are used to perform mathematical calculations such as addition, subtraction, multiplication, division, remainder, integer division, and power.

Example:

`a <- 10`

`b <- 3`

`a + b`

`a - b`

`a * b`

`a / b`

`a %% b`

`a %/% b`

`a ^ b`

## Output:

```
[1] 13
[1] 7
[1] 30
[1] 3.333333
[1] 1
[1] 3
[1] 1000
```

## 2. Assignment Operations

**Explanation:** Assignment operators are used to assign or store a value in a variable.

```
a <- 10
```

```
b = 20
```

```
30 -> c
```

```
print(a)
```

```
print(b)
```

```
print(c)
```

## Output:

```
[1] 10
[1] 20
[1] 30
```

### 3. Comparison Operations

**Explanation:** Comparison operators are used to compare two values. The result is TRUE or FALSE.

Example:

`a <- 10`

`b <- 20`

`a == b`

`a != b`

`a > b`

`a < b`

`a >= b`

`a <= b`

**Output:**

```
[1] FALSE
[1] TRUE
[1] FALSE
[1] TRUE
[1] FALSE
[1] TRUE
```

## 4. Logical Operations

**Explanation:** Logical operators are used to combine or change logical conditions. The main operators are AND, OR, and NOT.

Example:

```
a <- TRUE
```

```
b <- FALSE
```

```
a & b
```

```
a | b
```

```
!a
```

**Output:**

```
[1] FALSE
[1] TRUE
[1] FALSE
```

## 5. Miscellaneous Operations

**Explanation:** Miscellaneous operators are used for special operations such as creating sequences, checking membership, and accessing elements.

Example:

```
x <- 1:5
```

```
x
```

```
2 %in% x
```

**Output:**

```
[1] 1 2 3 4 5  
[1] TRUE
```

## Important Operators

- Arithmetic: +, -, \*, /, %% , %/%, ^
- Assignment: <-, =, ->
- Comparison: ==, !=, >, <, >=, <=
- Logical: &, |, !
- Miscellaneous: :, %in%, \$, %\*%